

ER Diagram & Its Components:-

An Entity - Relationship (ER) diagram is a visual representation of database's logical structure. It illustrates how data is stored, the entities (objects) that exist in system, & how these entities relate to each other. ER diagrams are one of the most widely used tools in db design because they helped developers, database administrators & stakeholders understand the system's data model before implementation.

This concept was first introduced by Peter Chen in 1976 as part of Entity - Relationship Model. These diagrams act as a blueprint for designing relational database, ensuring that all necessary data elements & their relationships are clearly defined before any coding or physical implementation takes place.

Main components of ER diagram:-

- ① Entities:-
Represent real-world objects (eg:- user, Product, etc.).
 - Strong Entity :- Exists independently.
 - Weak Entity :- Depends on strong entity.

- ② Attributes:-
Properties of entities or relationships.
 - Simple, composite, Derived or Multivalued.